In [1]:
```
# Random Forest Classification

# Confusion Matrix

# Model report

# Evalution Matrics= Accuracy, Specificity, Sensitivity, Precision, Recall,
f1 score, Macro Average, Weighted Average
```

In [2]:
```
# importing libraries

import pandas as pd
import numpy as ny
import matplotlib.pyplot as plt
```

In [3]:
```
dataset=pd.read_csv("Social_Network_Ads.csv")
```

In [4]:
```
dataset
```

Out[4]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

In [5]:
```
dataset=pd.get_dummies(dataset,drop_first=True)
```

In [6]:
```
dataset
```

Out[6]:

	User ID	Age	EstimatedSalary	Purchased	Gender_Male
0	15624510	19	19000	0	1
1	15810944	35	20000	0	1
2	15668575	26	43000	0	0
3	15603246	27	57000	0	0
4	15804002	19	76000	0	1
...	...	...	...	...	...
395	15691863	46	41000	1	0
396	15706071	51	23000	1	1
397	15654296	50	20000	1	0
398	15755018	36	33000	0	1
399	15594041	49	36000	1	0

400 rows × 5 columns

In [7]:
```
dataset["Purchased"].value_counts()
```

Out[7]:
```
0    257
1    143
Name: Purchased, dtype: int64
```

In [8]:
```
indep=dataset[["Age","EstimatedSalary","Gender_Male"]]
dep=dataset[["Purchased"]]
```

In [9]:
```
indep.shape
```

Out[9]:
```
(400, 3)
```

In [10]:
```
dep
```

Out[10]:

	Purchased
0	0
1	0
2	0
3	0
4	0
...	...
395	1
396	1
397	1
398	0
399	1

400 rows × 1 columns

In [11]:
```
# Spliting into training and test set
from sklearn.model_selection import train_test_split
X_train, X_test, Y_train, Y_test= train_test_split(indep,dep,test_size=0.3
0,random_state=0)
```

In [12]:
```
from sklearn.ensemble import RandomForestClassifier
classifier= RandomForestClassifier(n_estimators=10,criterion='entropy', ran
dom_state=0)
classifier.fit(X_train,Y_train)
```

```
C:\Users\user\Anaconda3\lib\site-packages\ipykernel_launcher.py:3: DataCon
versionWarning: A column-vector y was passed when a 1d array was expected.
Please change the shape of y to (n_samples,), for example using ravel().
  This is separate from the ipykernel package so we can avoid doing import
s until
```

Out[12]:
```
RandomForestClassifier(bootstrap=True, class_weight=None, criterion='entro
py',
                       max_depth=None, max_features='auto', max_leaf_nodes
=None,
                       min_impurity_decrease=0.0, min_impurity_split=None,
                       min_samples_leaf=1, min_samples_split=2,
                       min_weight_fraction_leaf=0.0, n_estimators=10,
                       n_jobs=None, oob_score=False, random_state=0, verbo
se=0,
                       warm_start=False)
```

In [13]:
```
Y_pred=classifier.predict(X_test)
```

In [14]:
```
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(Y_test,Y_pred)
```

In [15]:
```
print(cm)
```

```
[[72  7]
 [ 6 35]]
```

In [16]:
```
from sklearn.metrics import classification_report
clf_report=classification_report(Y_test,Y_pred)
```

In [17]:
```
print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.92      0.91      0.92        79
           1       0.83      0.85      0.84        41

    accuracy                           0.89       120
   macro avg       0.88      0.88      0.88       120
weighted avg       0.89      0.89      0.89       120
```

In [ ]: